Can a bank's disclosed diversification effect be reproduced from public data?

Evidence from ten years of Goldman Sachs Value-at-Risk disclosures

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Abstract

Every large dealer reporting market risk under Item 7A publishes Value-at-Risk by risk category and a single reconciling line between those categories and the firmwide total. The category figures are levels. The reconciling line, the diversification effect, is a relationship, and the relationship it reports is the one thing the filing withholds. That asymmetry makes it the single disclosed quantity an outside party can check. This paper tests it on ten consecutive Goldman Sachs filings, fiscal years 2016 through 2025, with a historical-simulation engine run at the bank's own disclosed settings — full revaluation, a five-year scenario window and age-weighted scenarios — over a proxy book of seventeen public risk factors sized to the disclosed split of risk across categories.

The level reproduces and the steadiness does not. Goldman's effect has averaged 42.0% of summed category VaR with a year-to-year standard deviation of 2.08 percentage points; a book sized once and then left alone averages 41.6%, which no test on ten annual observations separates from the disclosure, with a standard deviation of 6.33 ($F = 9.26$, $p = 0.003$). The firm's second disclosed measure locates the failure. Period-end against period-end the two amplitudes cannot be told apart; year on year the disclosed series carries nothing forward (lag-1 autocorrelation −0.47) while the proxy carries almost everything (+0.9). What fails to reproduce is not how far the ratio moves but whether it comes back. Sweeping the undisclosed decay rate across sixteen values, from equal weighting to a 6.6-day half-life, never brings the proxy nearer than three times the disclosed steadiness, and the rates that best repair the engine's coverage are among the worst on steadiness. Across all sixteen sleeve-sign configurations, five match the disclosed steadiness and none of those averages above 35.6%, eight reach 40% and the steadiest of those varies by 3.68 points; none does both.

Keywords: Value-at-Risk; diversification effect; risk disclosure; historical simulation; backtesting.

JEL classification: G21, G28, G32.

1 Introduction

Every large dealer that reports market risk under Item 7A of Form 10-K publishes a table with roughly the same shape. Average daily Value-at-Risk is broken out by risk category, the categories are added up, and a single negative line is subtracted to arrive at the firmwide figure. Goldman Sachs calls that line the diversification effect, and defines it as the difference between total VaR and the sum of the VaRs of its four risk categories: interest rates, equity prices, currency rates and commodity prices. Every other entry in the table is a level. How much could be lost on rates, on equities, on currencies, on commodities. The diversification line is not a level. It is the only number in the table that depends on how those four exposures move against one another, and co-movement is the one input the filing never publishes.

That asymmetry is what makes the line testable from outside, and it is also what bounds any such test. A reader cannot check a disclosed interest-rate VaR without knowing the rate book. A reader who builds a portfolio consistent with the disclosed split of risk across categories, and exposes it to public market data, can at least ask whether the co-movement that portfolio generates resembles the co-movement the filing reports. That is the exercise here, and the portfolio is a proxy throughout. It is calibrated so that each sleeve carries the share of standalone risk that Goldman discloses for the corresponding category; it is not a reconstruction of Goldman's positions, and its dollar size is a normalisation that carries no information. What is being tested is whether a static book with the disclosed risk structure, driven by public factors, produces the disclosed diversification. Nothing in the paper identifies a position, a desk, or a hedge.

The disclosures themselves make the test worth running, and they are more specific about method than the VaR tables alone suggest. Jorion (2002) showed that bank VaR disclosures are informative about subsequent trading volatility, which is the premise any replication presumes. The literature that followed has concentrated on the quality and the conservatism of the reported numbers: Pérignon and Smith (2010b) graded disclosure practice across a cross-section of commercial banks, Pérignon, Deng, and Wang (2008) asked whether the levels banks report are systematically too high, and Pérignon and Smith (2010a) examined the diversification component directly, on quarterly observations for four US banks running to the first quarter of 2007. Those studies work across institutions, and at the reporting frequency their data allow.

This paper adds three things to that base. The first is a single-dealer series long enough to contain regime breaks rather than straddle one. Goldman's diversification effect is reconstructed from ten consecutive filings covering fiscal years 2016 through 2025, a decade that includes the March 2020 dislocation and the 2022–2023 tightening cycle. The second is a replication run at the bank's own disclosed settings. Every filing in the sample states that Goldman computes VaR by historical simulation with full valuation, samples its scenarios from five years of historical data, and weights that data so that the importance of older observations decays. All three are implemented here, on seventeen factors drawn from the US Treasury and ECB curves and from exchange-traded prices, with the interest-rate sleeve valued by full bond revaluation rather than by a duration approximation. The engine is therefore capable of disagreeing with the disclosure in a way that an identity built out of the disclosed numbers cannot, and when it does disagree the disagreement cannot be attributed to a cruder method. The third is a decomposition. Goldman discloses the same quantity twice, as an annual average and as a period-end snapshot, and comparing each against its proxy analogue separates how far the ratio moves from whether those movements persist. The separation is what turns a failure to reproduce stability into a specific claim about mean reversion, and it uses only what the filings already publish. Running the complete space of sign configurations for the four sleeves then bounds how closely a static book can match the level and the steadiness at once, which turns out to be the binding constraint.

The disclosed series is remarkably flat. Across the ten years the diversification effect ranges from 39.0% to 45.5% of summed category VaR, with a mean of 42.0% and a standard deviation of 2.08 percentage points. A regression on the year index finds no trend ($p = 0.23$). A decade containing a pandemic, a zero-rate period and more than 500 basis points of policy tightening moved the reported co-movement of Goldman's four risk categories by about two points of standard deviation, and moved it in no particular direction.

The replication reproduces that level and does not reproduce that steadiness. At the disclosed five-year window, a book sized once on 2019 and then left alone averages 41.6% against the disclosed 42.0%; the two levels are statistically indistinguishable ($p = 0.84$). Its standard deviation across those years is 6.33 percentage points against the disclosed 2.08, a difference a variance-ratio test puts at $p = 0.0028$. And the two move differently over the decade: Goldman's second-half mean sits 0.8 points above its first, while the proxy's sits 8.5 points below.

Goldman's second disclosed measure of the same quantity says what kind of failure it is. Alongside the average daily figure the filings give a period-end snapshot, and that series has a standard deviation of 6.72 points: the firm's diversification ratio moves a great deal, and what is steady is its mean over a year rather than the ratio itself. Measured period-end against period-end the proxy's variation is statistically indistinguishable from Goldman's. What separates them is persistence. The lag-1 autocorrelation of consecutive annual values is -0.47 for Goldman's average daily series and essentially zero for its period-end series, against roughly +0.9 for both proxy series: Goldman's movements do not carry forward and the proxy's do. What fails to transfer is therefore not stability but mean reversion, and mean reversion is what the disclosed steadiness consists of.

The natural objection is that a proxy book on seventeen factors is simply a blunter instrument than the bank's, and that the gap measures the crudeness of the replication. Adopting Goldman's disclosed age weighting tests that objection directly, and refutes it. Weighting the scenario set toward recent observations raises the year-to-year standard deviation from 6.33 to between 9.46 and 10.04 points, four to five times the disclosed figure, across decay rates with half-lives of three weeks and three months. It does so while making the engine a demonstrably better risk model: the exception rate falls from 6.98% to 5.45% against a nominal 5%, and the Kupiec proportion-of-failures test moves from rejecting the model below any conventional level to a p-value of 0.328. The disclosed weighting is the correct specification, the filings say so, and applying it widens the stability gap rather than closing it.

Nor is the gap an artefact of one calibration of the sleeves. The four can be signed in sixteen ways and all sixteen were run. Taking the disclosed standard deviation itself as the bar, five configurations are at least that steady and the highest of those averages 35.64%, six points below the disclosure; eight average 40% or more and the steadiest of those varies by 3.68 points a year. No configuration achieves both. Within this engine the two properties of the disclosed series trade off against each other, and that frontier, rather than any single calibration, is the paper's central result. It also points at what a static book omits. A dealer that reprices positions daily, and whose hedges respond to the correlations it observes, can hold a diversification ratio steady in a way that a fixed portfolio exposed to the same market cannot. Goldman's own filings attribute its year-over-year VaR changes to exposure movements as well as to volatility in four consecutive years.

The remainder of the paper is organised as follows. Section 2 sets out what Goldman discloses, how the series was extracted from the filings, and where this exercise sits relative to prior work. Section 3 describes the factor panel and the choices that shaped it. Section 4 specifies the historical-simulation engine, the disclosed settings it adopts, and the calibration that ties the proxy book to the disclosed split. Section 5 reports the validation: unit-level checks, independent reproduction of the backtest statistics, and the formal coverage tests applied to the engine's own output. Section 6 presents the comparison against the disclosures and Section 7 isolates which assumptions the stability result depends on. Section 8 states what the exercise cannot establish, and Section 9 concludes. Code, extracted disclosures and model output are available in the repository cited in the data availability statement.

2 The disclosed diversification effect, and prior work

2.1 What Goldman discloses, and how the effect is defined

The definition is Goldman's own, and it is worth quoting because the whole exercise depends on it. In the words of the FY2025 filing, the "[d]iversification effect in the tables below represents the difference between total VaR and the sum of the VaR for the four risk categories," and "[t]his effect arises because the four market risk categories are not perfectly correlated" (The Goldman Sachs Group, Inc., 2017–2026). The four

Table 1. Goldman Sachs average daily Value-at-Risk by risk category, and the disclosed diversification effect, FY2016–FY2025

	Risk category							
Year	Rates	Equities	FX	Commod.	Summed	Diversif.	Total	% of sum
2016	45	25	21	17	108	-45	63	41.7
2017	40	24	12	13	89	-35	54	39.3
2018	46	31	14	11	102	-42	60	41.2
2019	46	27	11	12	96	-40	56	41.7
2020	71	55	23	20	169	-75	94	44.4
2021	60	43	13	25	141	-55	86	39.0
2022	97	33	32	47	209	-95	114	45.5
2023	96	29	24	19	168	-69	99	41.1
2024	81	37	26	19	163	-71	92	43.6
2025	70	47	24	17	158	-68	90	43.0

In millions of US dollars, except the final column. Average daily 95% one-day Value-at-Risk as reported in the Market Risk Management section of each fiscal year's Form 10-K, taken from the filing that first reported the year rather than from a later filing's restatement. *Summed* is the sum of the four category figures and is not itself disclosed. The diversification effect is disclosed as a negative number and reproduced with its sign; the final column expresses its magnitude as a percentage of the summed categories. Total VaR equals the summed categories plus the diversification effect in every year, exactly. The percentage averages 42.0 with a standard deviation of 2.08 across the ten years.

categories are interest rates, equity prices, currency rates and commodity prices. The measure is a one-day 95% VaR, and the filings state that a single model produces both the category figures and the total, which matters here: the diversification effect is internal to one model run rather than the residual of aggregating four separate ones.

Table 1 gives the series. Across the ten fiscal years the effect has sat between 39.0% and 45.5% of summed category VaR, averaging 42.0% with a standard deviation of 2.08 percentage points and no discernible trend. The dollar figures underneath it move a great deal: summed category VaR runs from $89 million in 2017 to $209 million in 2022, and total VaR from $54 million to $114 million over the same two years. The ratio of the two barely moves.

The filings are also more specific about method than the tables alone suggest, and three of those specifics are load-bearing later. Goldman states that it uses "historical simulations with full valuation of market factors at the position level by simultaneously shocking the relevant market factors for that position"; that it samples "from five years of historical data to generate the scenarios"; and that "[t]he historical data is weighted so that the relative importance of the data reduces over time," which "gives greater importance to more recent observations and reflects current asset volatilities" (The Goldman Sachs Group, Inc., 2017–2026). The FY2016 and FY2017 filings put the factor count at approximately 70,000; later filings drop the number. Section 4 adopts the method, the window and the weighting scheme; the decay rate is the one piece of the description the filings leave unquantified.

One consequence of the weighting is stated in the filings explicitly: "even if our positions included in VaR were unchanged, our VaR would increase with increasing market volatility and vice versa" (The Goldman Sachs Group, Inc., 2017–2026). The disclosed measure is therefore expected to respond to volatility without any change in exposure, which is the behaviour a static replication can in principle reproduce. What a static

replication cannot reproduce is a change in exposure, and Section 6 shows that the filings attribute much of the year-over-year movement to exactly that.

2.2 Where the number comes from

The figures are read out of the Market Risk Management section of each Form 10-K, filed under CIK 0000886982, for fiscal years 2016 through 2025. None of it is available as structured data: the Item 7A tables carry no XBRL tagging, so the numbers exist only as HTML table cells, and the cell layout is not stable across filings. Six tables are extracted from each filing where present: average daily VaR by category, period-end VaR, the year's high and low by category, the daily net revenue distribution, and the exception counts at the 95% risk-management measure and the 99% regulatory one. Appendix A gives the full extracted series and the accession number of every filing.

Two features of the disclosures do the work of validating that extraction.

The first is an identity. Total VaR must equal the sum of the four categories plus the diversification effect, because that is how the effect is defined. It is therefore a property of a correct parse rather than a finding, and it is a sharp one: a misread digit, a dropped minus sign or a column misalignment breaks it. Across every average-daily row extracted — 22 rows, since each filing restates the prior year, covering 12 distinct disclosure years from 2014 to 2025 — the residual is exactly zero. Net revenue bucket counts likewise sum to the stated annual totals in all fourteen year-filing pairs that report them.

The second is redundancy. Because each filing restates the preceding year, most years appear twice under different source filings, and the two readings agree everywhere but once. The FY2023 filing reports 2022 average daily VaR as $113 million total against $210 million summed, where the FY2022 filing had reported $114 million against $209 million. The filing explains the change: during the first quarter of 2023 Goldman added the currency exposure on certain debt and equity positions to VaR, removed certain other positions, and conformed the prior period. Both readings satisfy the identity, which is what identifies this as a change in the measure's scope rather than a correction of an error. Figures used throughout this paper are as first reported, so that a methodology change does not sit inside the time series.

Disclosure practice is not perfectly stable either. The 99% regulatory exception count appears in eight of the ten filings but not in the FY2022 or FY2023 filings, which report the 95% count alone. Nothing in this paper depends on the missing rows, but they are a reminder that what can be reconstructed from disclosure is determined by the filer rather than the researcher.

2.3 Prior work

The premise that these disclosures carry information is Jorion (2002)'s: reported VaR predicts the variability of subsequent trading revenue. What followed has largely asked how good the reported numbers are.

Two findings recur. The first is conservatism. Pérignon et al. (2008) study six Canadian banks on daily data from 1999 to 2005 and find systematic overstatement, two exceptions in 7,354 trading days against 74 expected at 99%, which they attribute partly to banks underestimating the diversification effect when aggregating across risk categories. Pérignon and Smith (2010b) construct a disclosure-quality index over 1996 to 2005 and report that historical simulation is by far the most common method in practice, and that VaR computed that way carries very little information about future volatility. The second is that bank VaR adapts slowly. O'Brien and Szerszeń (2017) find that exceedances at large US banks were excessive and clustered during the financial crisis, that a GARCH-based benchmark achieved lower exceedance rates without

clustering, and that variance decompositions show limited ability of bank VaR methodologies to adjust to crisis conditions. Campbell and Smith (2022) report the same shape in Australia, with VaR too high in quiet periods and too low in stressed ones.

Pérignon and Smith (2010a) is the nearest neighbour to this paper. It tests whether banks underaccount for diversification across broad risk categories, on quarterly observations for four US banks running to the first quarter of 2007, and finds no systematic underestimation: disclosed diversification effects are close to, and often larger than, the authors' own estimates. The level result in Section 6 corroborates that on a different bank, a different period and a different method, and it should be read as corroboration rather than discovery. Its sample ends before the financial crisis and is quarterly, which leaves the behaviour of a disclosed diversification effect across regime breaks as something a longer single-dealer series can speak to.

Daníelsson, James, Valenzuela, and Zer (2016) supply the mechanism this paper's central result turns on. They show that model risk, the potential for different models to give inconsistent outcomes, rises with market uncertainty: candidate models agree in calm periods and diverge significantly in distress. Their object is risk forecasts in general rather than a disclosed diversification ratio, but the diversification effect is a function of a correlation structure estimated from data, and the finding here is a reading of their result through that one statistic. Where their models diverge from each other, a reconstruction diverges from the thing it reconstructs.

Finally, the methodological literature this paper borrows from directly is small and is introduced where it is used: Kupiec (1995) and Christoffersen (1998) for the coverage tests, Basel Committee on Banking Supervision (1996) for the supervisory zones, Boudoukh, Richardson, and Whitelaw (1998) for the age-weighted quantile that implements Goldman's disclosed weighting, and Hull and White (1998) with J.P. Morgan and Reuters (1996) for the volatility-weighted variant.

3 Sample and data

3.1 Why one dealer over ten years

A cross-section of banks and a single bank over a long period answer different questions. A cross-section can establish how disclosure practice varies, which is what Pérignon and Smith (2010b) and Campbell and Smith (2022) do. A long single-dealer series can ask whether one firm's reported co-movement holds up across a change of regime, which needs the regime change inside the sample rather than at its edge.

That is the trade this paper makes. One filer, ten consecutive fiscal years, and enough of them that the sample contains the March 2020 dislocation, the zero-rate period that followed, and the tightening cycle of 2022 and 2023. The cost is that nothing here generalises to other banks: Goldman's disclosed diversification effect is a fact about Goldman, and the stability this paper fails to reproduce may be particular to its book. What is tested is a reproduction, not a cross-sectional regularity.

3.2 The sample, and where its boundaries come from

The disclosures are the ten Form 10-K filings of The Goldman Sachs Group, Inc., CIK 0000886982, covering fiscal years 2016 through 2025 and filed between February 2017 and February 2026. Appendix A lists the accession number, filing date and retrieval URL of each.

Both sample boundaries follow from one rule, stated in Section 2: every figure is taken from the filing that first reported it, rather than from a later filing's restatement of it. Each 10-K restates the prior year,

so the FY2016 filing does carry 2014 and 2015 figures, but only as comparatives. Reaching those years as first reported would require the FY2014 and FY2015 filings, which are not in the sample because the ten-year window was fixed before the as-first-reported rule made their absence binding. Extending the series backwards is therefore a matter of fetching two more filings rather than a limitation of method, and it was not done. At the recent end, FY2025 is simply the most recent filing available at the retrieval date of 9 October 2026.

The factor panel's start date is set by the disclosed window rather than chosen. A five-year lookback needs five years of scenarios before the first valuation date, and the aligned trading calendar described below discards about 9% of days, so five years of scenarios needs closer to six calendar years of history. The panel begins on 5 January 2010 and runs to 30 December 2025, 3,670 aligned days, which leaves 1,376 days of history before the first valuation date against the 1,260 the window consumes. One restatement in the disclosed series is discussed in Section 2 and is not repeated here; the figures used throughout are as first reported.

3.3 The seventeen factors

Table 2 lists the risk factors and Table 3 summarises their daily moves. All seventeen come from the issuer of the instrument, a central bank, or an exchange close: the US Treasury par yield curve at two, five, ten and thirty years (U.S. Department of the Treasury, 2026); the European Central Bank's euro area AAA central government spot curve at two, five and ten years (European Central Bank, 2026); four equity indices; three currency pairs; and three commodity funds. They are grouped to match Goldman's four disclosed categories so that each sleeve can be sized against the corresponding disclosed share.

Seventeen factors against the roughly 70,000 the FY2016 and FY2017 filings describe is the single largest difference between this model and Goldman's, and the direction of its effect is known: fewer independent sources of risk means less diversification available, which biases the measured effect downward relative to a book with a richer factor set. It is not quantifiable here, and Section 8 says so rather than estimating it.

The two blocks of Table 3 are in different units and are not comparable with each other. That is not a presentational awkwardness but the reason the engine sizes sleeves against disclosed risk shares rather than against notional: a dollar of two-year Treasury and a dollar of crude oil fund carry risks that differ by more than an order of magnitude, so matching Goldman's category proportions requires matching risk, not money. Section 4 gives the calibration.

3.4 Three choices that shaped the panel

Each of these changes the numbers, so each is stated rather than left in the code.

Equity indices enter in local currency. A dollar investor holding the Nikkei carries yen risk as well as Japanese equity risk. Goldman reports currency risk as a category of its own, so converting the indices to dollars would push currency risk into the equity sleeve and count it twice, making the category attribution — the thing being matched — meaningless.

Commodities enter through exchange-traded funds rather than front-month futures. On 20 April 2020 the NYMEX May 2020 WTI contract settled at −$37.63 per barrel, against a close of $18.27 on 17 April (U.S. Commodity Futures Trading Commission, 2020). A percentage return is not defined across a sign change, and the implied one-day move of −306% is arithmetic rather than a price movement a risk model should be asked to price. Funds cannot settle below zero. The cost of the substitution is roll yield, which drags a fund

Table 2. The seventeen risk factors, grouped to match Goldman's four disclosed risk categories

Factor	Category	Instrument	Unit of daily move	Source
us_2y	Interest rates	Par coupon bond, semiannual	Yield change (pp)	US Treasury
us_5y	Interest rates	Par coupon bond, semiannual	Yield change (pp)	US Treasury
us_10y	Interest rates	Par coupon bond, semiannual	Yield change (pp)	US Treasury
us_30y	Interest rates	Par coupon bond, semiannual	Yield change (pp)	US Treasury
eur_2y	Interest rates	Zero-coupon bond, annual	Yield change (pp)	ECB
eur_5y	Interest rates	Zero-coupon bond, annual	Yield change (pp)	ECB
eur_10y	Interest rates	Zero-coupon bond, annual	Yield change (pp)	ECB
equity_us	Equity prices	S&P 500 index	Fractional return	Exchange
equity_europe	Equity prices	EURO STOXX 50 index	Fractional return	Exchange
equity_japan	Equity prices	Nikkei 225 index	Fractional return	Exchange
equity_em	Equity prices	Emerging-markets equity fund	Fractional return	Exchange
fx_eurusd	Currency rates	EUR/USD spot	Fractional return	Exchange
fx_usdjpy	Currency rates	USD/JPY spot	Fractional return	Exchange
fx_gbpusd	Currency rates	GBP/USD spot	Fractional return	Exchange
commodity_gold	Commodity prices	Gold fund	Fractional return	Exchange
commodity_oil	Commodity prices	Crude oil fund	Fractional return	Exchange
commodity_base_metals	Commodity prices	Base metals fund	Fractional return	Exchange

Rate factors are repriced by full revaluation, which needs the level as well as the change, because the same 25 basis point move is worth a different amount of money at a 0.5% yield than at a 5% one. A yield change of 0.25 means 25 basis points throughout. Equity indices enter in local currency, so that currency risk stays in the currency sleeve rather than being counted twice. Commodities enter through exchange-traded funds rather than front-month futures; Section 3 gives the reason.

away from spot over long horizons; over one day it is second order, but it is a real difference from a spot book and it is carried through to Section 8.

Prices are adjusted for splits. The crude oil fund used here ran a one-for-eight reverse share split, effective after the close on 28 April 2020, with post-split shares trading from the following day (United States Oil Fund, LP, 2020). On unadjusted closes that is a 700% overnight gain, and it would enter the scenario set as a tail event in the middle of the most consequential stress window in the sample, within days of the negative settlement above. Split-adjusted closes are used throughout, for every traded instrument rather than only this one.

3.5 The trading calendar

Tokyo, London, New York and the ECB keep different holidays, and the panel is the intersection of their calendars rather than the union. Taking only days on which every market traded costs about 9% of trading days and leaves 225 to 233 observations a year against the 250 to 253 Goldman reports. The alternative, forward-filling a closed market, would plant runs of exact zeros in the scenario set and understate that factor's risk, which is a worse error than a smaller sample: it would bias the thing being measured rather than merely reducing the precision with which it is measured.

The reduced sample does cost power in every test reported in Section 5, and that is the price of the choice.

Two further properties of the panel are limitations rather than choices, and neither is corrected. The surviving closes are still not simultaneous across time zones, which depresses measured cross-market

Table 3. Daily moves by factor, 2010–2025

Factor	Mean	Std. dev.	Minimum	Maximum
us_2y	+0.06	4.73	-57.00	+34.00
us_5y	+0.03	5.62	-35.00	+31.00
us_10y	+0.01	5.63	-36.00	+29.00
us_30y	+0.00	5.51	-45.00	+29.00
eur_2y	+0.02	3.80	-43.47	+36.57
eur_5y	-0.01	4.35	-31.25	+29.35
eur_10y	-0.02	4.55	-26.46	+24.01
equity_us	+0.056	1.134	-11.984	+9.515
equity_europe	+0.026	1.311	-12.401	+10.348
equity_japan	+0.052	1.381	-12.396	+10.226
equity_em	+0.026	1.409	-12.479	+8.053
fx_eurusd	-0.004	0.576	-5.001	+3.176
fx_usdjpy	+0.016	0.621	-4.552	+4.061
fx_gbpusd	-0.003	0.586	-7.604	+3.791
commodity_gold	+0.041	1.043	-8.781	+5.984
commodity_oil	-0.014	2.325	-25.315	+16.667
commodity_base_metals	+0.014	1.283	-8.371	+6.191

Equity, currency and commodity factors in percent per day; interest-rate factors in basis points per day. The two blocks are not comparable with one another, which is why the engine sizes sleeves against disclosed risk shares rather than against notional. 3,670 aligned trading days from 5 January 2010 to 30 December 2025.

correlation and therefore overstates diversification, in the opposite direction to the factor-count bias above, which is why neither is used to argue the result one way or the other. And the Treasury curve is quoted to the basis point, so at near-zero policy rates the two-year yield shows no change at all on 21.1% of days across the panel, rising to 43.1% of days in the second half of 2020. The build's staleness check accommodates this by allowing longer runs of identical values in yield factors than in traded prices; the longest unbroken run in the two-year series is six days, against a limit of fifteen. The discretisation slightly understates true rate variability through the near-zero years.

4 Method

4.1 Instruments, and what full revaluation means here

Equity, currency and commodity positions are linear in their factor: a one-day fractional return on the factor is the fractional profit and loss on the position. Interest-rate positions are not. They are repriced by discounting their cashflows again at the shocked yield, which is what Goldman's phrase "full valuation" describes, rather than multiplied by a modified duration.

The two rate sub-sleeves keep their source's native convention. US positions are par coupon bonds priced off the Treasury par yield curve with semiannual coupons; euro positions are zero-coupon bonds priced off the ECB spot curve with annual compounding. The alternative would be to convert one curve into the other's convention, which needs a bootstrap the published data does not support at these tenors. The cost of keeping both is that a dollar of euro zero and a dollar of US par bond are not interchangeable: a zero has longer

duration than a par bond of the same maturity, so the sleeves differ per unit of notional. The sizing described below targets risk rather than notional, which absorbs the difference, but the instrument sets are not identical and Section 8 records that.

Writing $P(y, m)$ for the price of the relevant instrument at yield y and maturity m, the scenario return for a rate position under a historical yield change Δy_s observed on day s is

$$r_s = \frac{P(y_t + \Delta y_s,\, m)}{P(y_t,\, m)} - 1, \tag{1}$$

where y_t is the yield at the close of the valuation date. The level enters as well as the change, which is why the engine carries both: the same 25 basis point move is worth a different amount of money at a 0.5% yield than at a 5% one. Equation (1) is also where convexity enters, and Section 7 reports what it is worth by running the same scenarios through the first-order approximation $r_s \approx -D_{\text{mod}}(y_t, m)\,\Delta y_s$ instead.

4.2 Sizing within a sleeve, and sizing the sleeve against the disclosure

Sizing happens in two stages, and only the second touches the disclosures.

Within a sleeve, instruments are weighted so that each contributes equal standalone VaR over the calibration window. This is a neutrality choice rather than an empirical claim: nothing is disclosed about the composition of Goldman's rate book, so loading the sleeve onto any one tenor would be an assumption with no support. Equal risk contribution is the weighting that requires no such assumption.

The sleeve as a whole is then scaled so that its average daily VaR across the calibration year equals the figure Goldman disclosed for that category. The scaling is available in closed form, which is worth stating because it is the design of the experiment rather than a computational convenience. Profit and loss on every position here is proportional to its notional, so for a sleeve c with notionals $\mathbf{n}$ scaled by $k > 0$,

$$\text{VaR}_c(k\mathbf{n}) = k\ \text{VaR}_c(\mathbf{n}), \tag{2}$$

and the same proportionality carries through the average over the year. One division per sleeve therefore lands exactly on the disclosed target. Section 5 confirms that it does, to machine precision.

Equation (2) is positive homogeneity, which is one of the four axioms Artzner, Delbaen, Eber, and Heath (1999) use to define a coherent risk measure and one of the three VaR actually satisfies. It is worth naming because the axiom VaR fails — subadditivity — is the subject of this paper's central quantity, and the calibration leans on the one property that holds while the experiment measures the one that does not.

The consequence is that the summed category VaR matches the disclosures by construction and carries no information whatever. That is deliberate. Fixing the four category figures leaves total VaR and hence the diversification effect entirely free, because the total depends on how the four sleeves move together and nothing in the calibration constrains that. Comparing the free quantity against its disclosed counterpart is the experiment; the matched quantities are the control.

Table 4 gives the resulting book. Its dollar level is a normalisation and should not be read as an estimate of anything: only the ratios between sleeves carry meaning, and those come from the disclosures.

Table 4. The fixed proxy book after calibration to the FY2019 disclosed risk split

Factor	Category	Instrument	Maturity (y)	Notional ($mm)
us_2y	Interest rates	Par coupon bond	2	8,403
us_5y	Interest rates	Par coupon bond	5	2,467
us_10y	Interest rates	Par coupon bond	10	1,302
us_30y	Interest rates	Par coupon bond	30	553
eur_2y	Interest rates	Zero-coupon bond	2	18,162
eur_5y	Interest rates	Zero-coupon bond	5	3,941
eur_10y	Interest rates	Zero-coupon bond	10	1,427
equity_us	Equity prices	Linear in factor	—	604
equity_europe	Equity prices	Linear in factor	—	478
equity_japan	Equity prices	Linear in factor	—	444
equity_em	Equity prices	Linear in factor	—	442
fx_eurusd	Currency rates	Linear in factor	—	788
fx_usdjpy	Currency rates	Linear in factor	—	734
fx_gbpusd	Currency rates	Linear in factor	—	701
commodity_gold	Commodity prices	Linear in factor	—	443
commodity_oil	Commodity prices	Linear in factor	—	165
commodity_base_metals	Commodity prices	Linear in factor	—	341

Notionals are the output of the sizing described in Section 4, not an estimate of anything Goldman holds. Sleeve totals are $36,255mm of rates, $1,968mm of equity, $2,223mm of currency and $949mm of commodity exposure, $41,395mm in all. The rate sleeve dominates on notional and not on risk: a two-year note carries a fraction of the daily risk of a dollar of crude oil fund, which is the whole reason sizing targets risk rather than money. The dollar level is a normalisation and carries no information; only the ratios between sleeves do.

4.3 The engine and its timing convention

For a valuation date t, the scenario set is the W most recent daily factor moves up to and including t, applied to the book as it stands at the market levels of t. Each scenario produces one profit-and-loss outcome per sleeve and one for the book as a whole, and VaR is the loss at the α quantile of that distribution, reported as a positive number. Category VaR uses only that category's positions. Total VaR moves every factor together, historical day by historical day, which is what preserves the co-movement the paper is testing. The diversification effect is

$$D_t = \mathrm{VaR}_t^{\mathrm{total}} - \sum_{c \in C} \mathrm{VaR}_t^c, \qquad d_t = \frac{-D_t}{\sum_{c \in C} \mathrm{VaR}_t^c}, \tag{3}$$

following Goldman's definition exactly, with d_t the share reported throughout this paper.

The settings are the disclosed ones where the filings supply them: $W = 1{,}260$ trading days, which is the five years Goldman states it samples from, and $\alpha = 0.05$ for the 95% risk-management measure. The quantile is read off the finite scenario set by linear interpolation; the discrete order statistic is the conservative alternative and Section 7 reports the difference rather than assuming it away. Appendix B gives the reading conventions in full, together with the argument behind the closed-form calibration and the look-ahead probe.

The timing convention is stated explicitly because getting it wrong produces a model that looks better than it is. VaR dated t uses only the book, the levels and the scenarios available at the close of t, and it is compared against the profit and loss realised on $t + 1$. No quantity dated $t + 1$ enters the VaR dated t. This is Goldman's

own convention: the filings state that daily net revenues are compared with VaR calculated as of the end of the prior business day. Section 5 describes the test that enforces it.

4.4 Scenario age weighting

Goldman states that its historical data is weighted so that the importance of older observations decays. Implementing that replaces the equally weighted quantile with a weighted one, following the hybrid approach of Boudoukh et al. (1998). A scenario observed k trading days before the valuation date enters with weight

$$w_k = \frac{\lambda^k}{\sum_{j=0}^{W-1} \lambda^j}, \qquad \lambda \in (0, 1], \tag{4}$$

and VaR is read at the point where cumulative weight, rather than cumulative count, reaches α. At $\lambda = 1$ every weight is $1/W$ and the estimator returns to the equally weighted percentile.

Two properties of the implementation matter for interpreting Section 7. The first is that the collapse at $\lambda = 1$ is exact, in both the interpolated and the discrete reading, which Section 5 reports as a test: any difference between a weighted and an unweighted run is therefore the weighting and not a change of estimator. The second is that λ is not disclosed. The filings describe the scheme but not its rate, so the baseline keeps $\lambda = 1$, leaving no undisclosed parameter inside it, and the rate is swept instead, at values chosen for their half-lives rather than for how close they land to the disclosures.

4.5 Two books, and what each is for

Two books are run, and conflating them would be a mistake.

The *recalibrated* book is resized every year to that year's own disclosed split. Its exposure mix tracks the disclosures exactly, which removes the exposure mix from the comparison and leaves only the cross-category correlations in the public data. It is the right book for asking whether the diversification effect comes out where Goldman reports it. It is not a forecast, and could not be: sizing the 2022 book uses a figure published in February 2023.

The *fixed* book is sized once, on 2019, and then left alone. Nothing about it responds to anything disclosed afterwards, so the path its VaR takes is driven purely by what happened to volatility and correlation. It is the right book for the regime comparison, and the only defensible one for backtesting, because its notionals were set before any of the data it is scored on. 2019 is the base year because it is calm and precedes everything the backtest scores, which makes 2020 onward out of sample with respect to sizing.

4.6 The volatility-weighted variant

One variant is run alongside the disclosed specification, because it addresses a failure the disclosed specification has. Hull and White (1998) rescale each historical return by the ratio of today's conditional volatility to the volatility in force when the return was observed, so the scenario set is restated in current volatility units before the percentile is taken. Volatility is estimated by exponentially weighted variance with the J.P. Morgan and Reuters (1996) daily decay factor of 0.94, and the estimator is causal by construction: the forecast stamped t uses only moves up to t. A GARCH(1,1) alternative is refitted each January on the history available at that point.

Realised profit and loss always comes from the unscaled moves. What the book actually made on $t + 1$ is not a function of which model forecast it, and scaling the outcome as well as the scenarios would make the backtest circular.

Age weighting and volatility weighting are easy to confuse and do different things. Age weighting changes how much a scenario counts according to when it happened. Volatility weighting changes the size of the scenario itself according to the volatility regime it came from. Both turn out to repair the engine's coverage, and neither repairs the diversification result, for a reason Section 7 develops: both act on the scale of the scenario set, and the diversification effect is a function of its correlation structure.

5 Validation

A model that runs is not a model that is right. Every check in this section compares the engine against something outside it — a hand calculation, an independent implementation, a published boundary, an algebraic identity — so that agreement is evidence rather than self-report. The suite is 107 tests. What none of it establishes is the resemblance between the proxy book and Goldman's, and the last subsection says so plainly.

5.1 The pricer against hand calculation

The bond pricer is the one component where an error would be silent: a mispriced bond produces a plausible VaR rather than an exception. A ten-year bond with a 5% semiannual coupon valued at a 6% yield prices at 92.5612626, which the implementation reproduces to seven decimals, and it trades below par because its coupon is a point under its yield. A zero-coupon bond reproduces $100/1.03^{10}$ to twelve decimals at a 3% rate. A par bond prices at exactly 100 at its own yield, which is the identity that catches a coupon-frequency error.

Two further properties are checked because Equation (1) depends on them. The analytic modified duration agrees with a numerical derivative of the price function, and convexity has the correct sign in both directions: a yield fall gains more than duration predicts and a yield rise loses less. That second property is what Section 7 quantifies when it prices the rate sleeve both ways.

5.2 The sizing identity

Section 4 claims the sleeve calibration is exact rather than approximate, on the grounds that VaR is positively homogeneous in notional. Table 5 checks the claim instead of assuming it. Across all 40 sleeve-years the largest absolute sizing error is 4.26×10^{-14} percent of target, which is floating-point noise.

Two things make this worth a table rather than a footnote. The first is that an inexact calibration would quietly contaminate the comparison: if summed category VaR only approximately matched the disclosures, part of any gap in the diversification effect would be a sizing artefact rather than a correlation result. The second is that the check is independent of the mechanism. The engine is not told that scaling is linear; it is run, and the achieved sleeve VaR is measured and compared against target. Agreement to fourteen decimal places is then evidence for the homogeneity property, not a restatement of it.

A test in the suite makes the same point directly, by running the whole engine on a book and on the same book scaled by an arbitrary factor and confirming that every VaR scales by exactly that factor.

Table 5. Sizing check: the disclosed sleeve targets, and the largest error the calibration leaves behind

	Disclosed target ($mm)				Largest error
Year	Rates	Equity	Currency	Commodity	(% of target)
2016	45	25	21	17	4.26e-14
2017	40	24	12	13	1.78e-14
2018	46	31	14	11	1.61e-14
2019	46	27	11	12	3.23e-14
2020	71	55	23	20	3.09e-14
2021	60	43	13	25	3.30e-14
2022	97	33	32	47	3.02e-14
2023	96	29	24	19	2.96e-14
2024	81	37	26	19	3.84e-14
2025	70	47	24	17	3.02e-14

Targets are Goldman's disclosed average daily VaR by category, as first reported. The final column is the largest absolute sizing error across the four sleeves in that year, as a percentage of target. Across all 40 sleeve-years the largest error is 4.26e-14%, which is floating-point noise: the calibration is exact, as Equation (2) requires, and this table verifies that claim rather than reporting a convergence result.

5.3 The backtest statistics against published values

The coverage tests are standard and their implementations are checked against external values rather than against each other.

The Kupiec (1995) proportion-of-failures statistic reproduces an independent implementation on two published cases. With 1,043 observations and 57 failures at 95%, the published statistic is 0.46147 with a p-value of 0.49694, and this implementation returns 0.4614664, agreeing to the five decimal places the published figure carries. With 17 failures at 99% the published statistic is 3.51180 and this implementation returns 3.5118129, a difference of 1.3×10^{-5}; the published figure is not the correctly rounded five-decimal value, which is 3.51181, so the agreement is to within that tolerance rather than digit for digit. The p-value, 0.060933, matches exactly. Its analytic limits hold as well. Zero exceptions give $-2T \ln(1 - p)$, and a failure rate exactly equal to coverage gives zero. A Monte Carlo experiment under the null puts the empirical size at 4.3% against a nominal 5% at the seed the test fixes, and between 4.4% and 5.7% across three further seeds. The statistic is discrete, so its distribution cannot match a continuous chi-square exactly; size at the nominal level is the property that matters and the one that is tested.

The Christoffersen (1998) independence and conditional coverage statistics are checked against their own analytic cases, including the degenerate samples where a transition count is zero.

The Basel traffic-light boundaries deserve a note because they are easy to get wrong by borrowing them. They are rederived here from the binomial distribution, not hard-coded, and the derivation reproduces the published boundaries of five and ten for a 99% model over 250 trading days (Basel Committee on Banking Supervision, 1996). The reason the derivation matters is that those numbers are specific to 99%. A correctly calibrated 95% model is expected to break about twelve times in 250 days, and the rederivation puts its yellow and red thresholds at 18 and 27; carrying the published 99% boundaries across unchanged would place a well-specified 95% model permanently in the red zone. Every zone reported in this paper is computed at the confidence level it belongs to.

5.4 The look-ahead probe

Look-ahead is the failure mode that announces itself as success. Every other error in a VaR engine produces an implausible number; using tomorrow's data produces an unusually good backtest, which is easy to mistake for a working model.

The probe is therefore adversarial. Every factor move after a cut-off date is multiplied by 25, and the engine is re-run on the tampered panel. VaR stamped on or before the cut-off must be identical to what it was, because it is only permitted to see scenarios up to its own date. The realised profit-and-loss column is excluded from the comparison, since that column is supposed to change: it carries the next day's outcome, which is exactly the quantity the tampering alters.

A companion test pins the off-by-one that would make the backtest meaningless, confirming that the realised result recorded against date t is the book repriced on the move of $t + 1$ and not of t.

5.5 The subadditivity guard

One check needs explaining rather than asserting, because its natural form would be wrong.

The diversification effect in Equation (3) is almost always negative, and an engine that produced a positive one would normally be signalling a sign error. But VaR is not a subadditive risk measure. Artzner et al. (1999) define a coherent measure by four axioms: translation invariance, subadditivity, positive homogeneity and monotonicity. They show that VaR satisfies the first, third and fourth while failing the second. In their words, the use of VaR "does not encourage and, indeed, sometimes prohibits diversification." On a sufficiently awkward joint tail, a whole book really can show a larger loss quantile than the sum of its parts. That is a property of the measure rather than a defect in the code, and it is the reason expected shortfall exists.

So the guard is a tolerance, not an assertion. The engine records days on which the effect is positive and refuses the run only when they exceed one day in a hundred, which is the threshold separating the genuine phenomenon from a sign error. A test confirms the guard behaves that way in both directions: it tolerates a few such days and rejects many.

On every configuration reported in this paper the count is zero, 0 of 2,293 days on both the fixed and the recalibrated book. That is not an accident of this data. Artzner et al. (1999) note in a remark that under joint normality the quantiles do satisfy subadditivity as long as exceedance probabilities stay below one half, and a four-sleeve book of broad market factors at a 5% tail is close enough to that case for the failure not to bite. The guard exists for the configurations where it might, not because it ever fired.

5.6 The equal-weight collapse

One check exists specifically to make Section 7 interpretable. The age-weighted estimator of Equation (4) and the ordinary percentile are different pieces of code, and if they disagreed at $\lambda = 1$ then any difference between a weighted and an unweighted run would mix the weighting with a change of estimator.

They do not disagree. The plotting position in the weighted reading is chosen so that equal weights reduce exactly to the position the unweighted percentile interpolates against, and the discrete readings coincide likewise. The suite pins the identity to machine precision across four sample sizes and four quantiles, in both conventions. Whatever Section 7 reports about age weighting is therefore the weighting.

5.7 What the engine does not validate

Everything above establishes internal correctness: the engine computes what Section 4 says it computes, on the data Section 3 describes. Within that, two things are checked by the build rather than by the suite and should be named: the alignment of the factor panel to a common trading calendar, and the staleness and sign checks on the raw series. Those run on every rebuild and halt it on failure, but they are not unit tests and no test asserts that the calendar intersection is correct beyond what those checks cover. None of it establishes that the proxy book resembles Goldman's trading book, and no test could, because the positions are not disclosed and never will be.

This matters for how the results should be read. A validated engine on a proxy book can show that a particular structure of market risk, exposed to public factor data, does or does not generate a particular diversification effect. It cannot show that Goldman's book has that structure. The level agreement in Section 6 is therefore evidence about the proxy, and the steadiness gap is evidence about what a static proxy cannot do, not a measurement of anything inside the firm.

6 Results

6.1 The level

The level reproduces. Table 6 sets the disclosed series beside both proxy books. Over the ten years Goldman's diversification effect averages 42.0% of summed category VaR; the book sized once on 2019 and then left alone averages 41.6%. A Welch test does not separate the two ($t = -0.20$, $p = 0.84$), which is the appropriate way to report the agreement: the difference is small, but with ten annual observations and a proxy mean whose 95% interval runs from 37.1 to 46.1, the precision implied by quoting it to a decimal place is not there. The book resized each year to the disclosed split averages 44.7%, which overshoots and sits just outside the disclosed range of 39.0 to 45.5 on the high side.

That the fixed book matches more closely than the recalibrated one is worth a sentence, because the opposite would be the natural expectation. Recalibration forces each sleeve's standalone VaR onto the disclosed figure every year, which removes the exposure mix from the comparison; it does not remove it from the correlation structure, because changing sleeve sizes changes how the sleeves offset one another in the joint tail. The fixed book has no such annual intervention, and over a decade its level happens to land closer. Neither result should be over-read: both are within a few points of a figure the engine was never told, and the agreement is the finding rather than the difference between the two books.

This result corroborates rather than discovers. Pérignon and Smith (2010a) found that disclosed diversification effects sit close to independently estimated ones, on four US banks and quarterly data ending in early 2007. The same holds here, on a different bank, a different decade and a different method (public factors and a proxy book rather than their estimation), which is what makes it corroboration worth reporting: the finding survives a change of almost everything except the question.

6.2 The steadiness

The steadiness does not reproduce, and the failure is directional rather than noisy.

Goldman's series has a standard deviation of 2.08 percentage points across the ten years. The fixed book's is 6.33, three times as large, and the difference is not an artefact of small samples: a variance-ratio test gives

Table 6. The disclosed diversification effect against the proxy book, FY2016–FY2025

	Diversification effect (% of summed)			Total VaR
Year	Goldman	Resized yearly	Fixed since 2019	error (%)
2016	41.7	53.9	49.3	-21.0
2017	39.3	47.6	46.1	-13.7
2018	41.2	44.8	43.2	-6.1
2019	41.7	45.1	45.1	-5.8
2020	44.4	49.6	45.5	-9.4
2021	39.0	48.1	45.8	-14.9
2022	45.5	43.6	42.4	+3.3
2023	41.1	37.7	34.5	+5.7
2024	43.6	39.1	32.3	+8.0
2025	43.0	37.9	31.9	+9.1
Mean	42.0	44.7	41.6	
Std. dev.	2.08	5.35	6.33	

The diversification effect as a share of summed category VaR, by Goldman's own definition. *Resized yearly* is the book recalibrated to each year's disclosed split; *fixed since 2019* is sized once on 2019 and left alone. The final column is the error in the fixed book's total VaR against the disclosed total, which changes sign in 2022. The dollar level of the book is a normalisation, so that column measures the consequence of the diversification gap rather than a sizing error: Table 5 shows the sizing is exact.

$F = 9.26$ on $(9, 9)$ degrees of freedom with $p = 0.0028$, and Bartlett's test agrees. Levene's test, which is robust to departures from normality and the safer choice at this sample size, is borderline at $p = 0.055$. The parametric result is reported with the robust one rather than instead of it.

The two series also move differently over the decade. Goldman's mean for 2021–2025 is 0.8 points above its mean for 2016–2020; the fixed book's is 8.5 points below, and the book finishes the sample 17.4 points lower than it started. Figure 1 shows the three series together, and the shape is the result: a flat disclosed line and two proxy lines that track it until 2022 and then fall away from it.

No regression trend or p-value is quoted for these movements, and the reason is worth stating because it constrains what any study of this kind can claim. The proxy's annual values are computed from overlapping five-year scenario windows: with about 230 trading days a year inside a 1,260-day window, consecutive annual figures share roughly 82% of the data that produces them. Least-squares standard errors on ten such observations assume an independence the construction rules out, and would overstate the significance of any trend by a wide margin. The decline is large and visible; it is not something a significance test on this sample can certify.

The mechanism is visible in the final column of Table 6. The error in the fixed book's total VaR changes sign in 2022. Before then the proxy over-diversifies, understating total VaR by as much as 21.0%; afterwards it under-diversifies, overstating total VaR by up to 9.1%. The turn is the equity–bond correlation moving positive in the inflation regime, which removes the main diversifier available to a directional book. Goldman's ratio went *up* that year, to its sample maximum of 45.5%.

The decline afterwards is a property of a long window on a static book. Once 2022 enters a five-year scenario set it stays there until 2027, so the measured correlation structure keeps the inflation regime in view long after the market has left it. Appendix C cuts the same daily output into market regimes, where the effect is starkest: the measured diversification falls from 44.8% in the calm years to 32.1% in 2024–2025, on a book

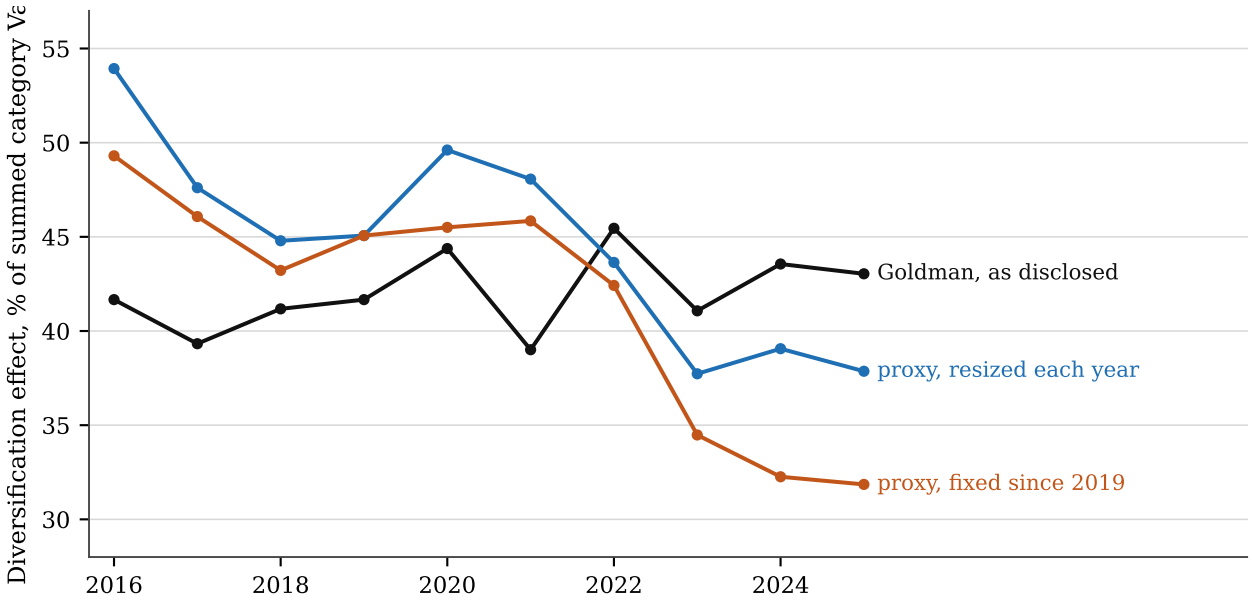


Figure 1. The disclosed diversification effect and the proxy book, FY2016–FY2025. Goldman's series is flat across a decade containing a pandemic, a zero-rate period and more than 500 basis points of policy tightening. Both proxy books track it until 2022 and then decline, as the five-year scenario window fills with the 2022 correlation regime and does not empty again.

that did not change. A dealer repositioning daily has no such ratchet, and Section 7 shows that shortening or reweighting the window does not rescue the result.

6.3 Amplitude against persistence

Goldman publishes the diversification effect twice: averaged over the year's daily values, and as a period-end snapshot. Everything above compares the average daily series, which is the measure the firm leads with. Comparing the other one separates two things the first comparison conflates — how far the ratio moves, and whether those movements persist — and the separation is informative enough to change how the gap should be described.

Table 7 sets both disclosed measures beside their proxy analogues. The first thing it shows is that Goldman's ratio is not a stable quantity. Measured period-end it has a standard deviation of 6.72 percentage points and a range of 28.6 to 48.6, against 2.08 and 39.0 to 45.5 for the average daily series. What is steady in the disclosures is the annual mean, not the ratio.

The decisive column is the last one. The lag-1 autocorrelation of a series, the correlation between consecutive annual values, distinguishes a quantity that returns toward its mean from one that carries its level forward, and it requires no model and no trend. Neither disclosed series shows positive persistence: the average daily series is −0.47 and the period-end series is essentially zero at +0.07. Both proxy series sit near +0.9. Whatever each series does in a given year, Goldman's does not carry forward and the proxy's does.

That is the finding, and it is sharper than a comparison of spreads. The proxy does not fail to vary, nor vary too much; it fails to come back. Reading Figure 1 again with that in mind, the proxy lines do not simply wobble more than the disclosed line; they leave and stay left.

The spread comparison corroborates this without carrying it. Measured period-end against period-end the proxy's standard deviation is 6.68 against Goldman's 6.72, which on inspection looks like an exact match

Table 7. Amplitude and persistence: the two disclosed measures of the diversification effect against their proxy analogues

Series	Mean	Std. dev.	Range	Change, first half to second	Lag-1 autocorr.
Goldman, average daily	42.03	2.08	39.0–45.5	+0.8	-0.47
Goldman, period-end	38.21	6.72	28.6–48.6	+8.6	+0.07
Proxy, average daily	41.60	6.33	31.9–49.3	-8.5	+0.88
Proxy, final day of year	40.62	6.68	30.6–47.8	-10.2	+0.90

All figures are the diversification effect as a percentage of summed category VaR across the ten fiscal years, so each column rests on ten observations and none is precisely estimated. *Change* is the mean of 2021–2025 less the mean of 2016–2020. *Lag-1 autocorr.* is the correlation between consecutive annual values: negative indicates a series that returns toward its mean, positive one that carries its level forward. Neither disclosed series shows positive persistence — the average daily series is −0.47 and the period-end series is essentially zero — while both proxy series sit near +0.9. That contrast, rather than any difference in spread, is what Section 6 rests on. No trend test is reported: the proxy's annual values are computed from overlapping five-year windows and share about 82% of their scenario set with their neighbours, so a regression on them would understate its own standard errors.

and on examination is not: each is estimated from ten observations, and the 95% confidence interval on the *ratio* of the two runs from 0.50 to 2.02. The defensible statement is that the two amplitudes are statistically indistinguishable ($F = 1.01$, $p = 0.98$), which is weaker than reproduction and is not evidence of equality. What the comparison does establish is that the gap between the two average daily series is not simply a matter of the proxy being more volatile, because measured alike the volatilities cannot be told apart.

Two further qualifications. Each period-end figure is a single trading day, so those rows rest on ten single-day observations and year-end snapshots may carry balance-sheet effects an average does not. And the proxy's final trading day is the last day of the aligned calendar rather than Goldman's fiscal period end.

6.4 The configuration space

A single calibration proves little, because the direction of each sleeve is not disclosed and the diversification effect depends on those directions more than on anything else in the specification. All sixteen sign combinations were therefore run on the fixed book, and Figure 2 plots each one's decade-average level against its year-to-year standard deviation.

Taking the disclosed standard deviation itself as the bar rather than a round number chosen afterwards: five of the sixteen configurations are at least as steady as Goldman's 2.08, and the highest of those averages 35.64%, six points below the disclosure. Eight average 40% or more, and the steadiest of those varies by 3.68 points a year, about 77% more than Goldman. No configuration achieves both.

This is a descriptive bound on what the engine can produce, and it should be read as one rather than as a test. Every standard deviation in Figure 2 is estimated from ten annual observations and carries a confidence interval spanning roughly a factor of two and a half in each direction; the steadiest configuration's 3.68 and Goldman's 2.08 are not separated by anything a formal comparison would certify. The sixteen configurations are not independent draws either, since they share the same factor data and differ only in sleeve signs, and the figures quoted are the selected extremes across them, with no adjustment for that selection.

What the bound does establish is the shape of the trade-off. Within this engine, moving a configuration toward Goldman's level moves it away from Goldman's steadiness, across the whole sign space rather than at one point. That is what makes the gap hard to explain away as a bad choice of signs: the better choice would have to lie outside the space, and the space is complete.

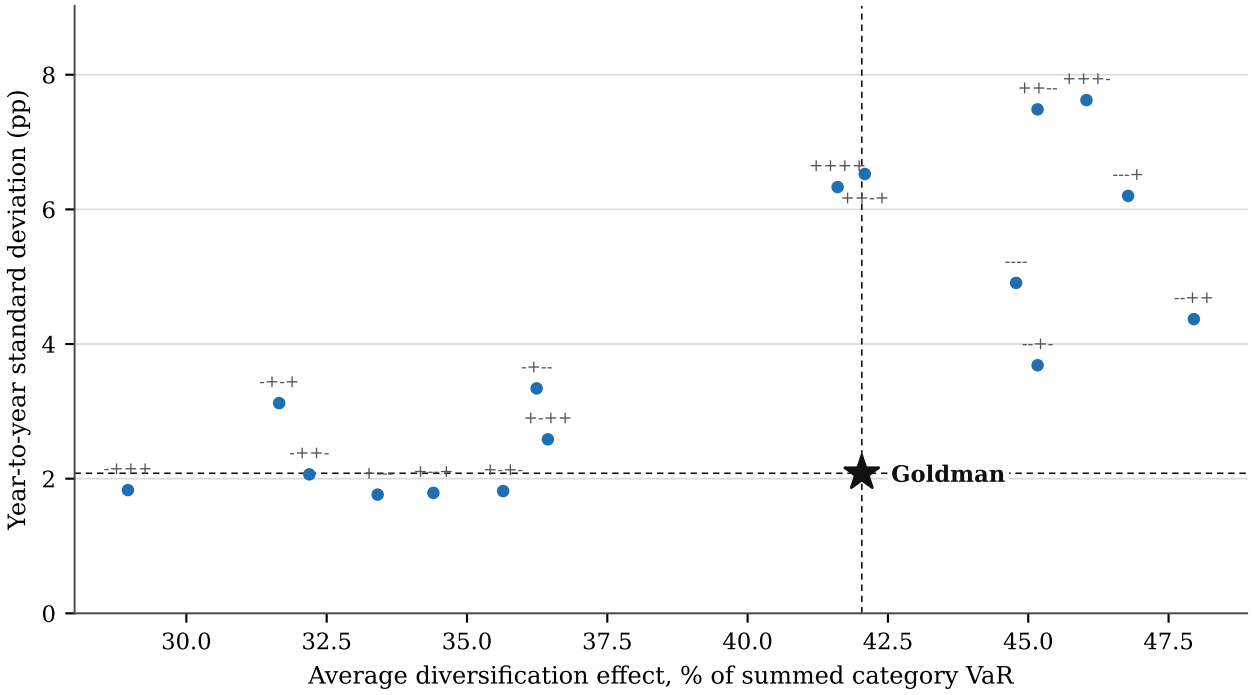


Figure 2. Level against steadiness for all sixteen sleeve-sign configurations, with Goldman's own position marked. Signs are in the order interest rates, equity, currency, commodity. The dashed lines are Goldman's average level and its year-to-year standard deviation; the quadrant below and to the right of their intersection — at least as steady as the disclosure, and at least as high — contains no configuration.

Table 8. Backtest results on the fixed book

Model	Conf.	Period	Days	Exc.	Rate (%)	Kupiec	Indep.	Joint	Zone
Historical simulation	95%	2016-2025	2,293	160	6.98	<0.0001	<0.0001	<0.0001	red
Historical simulation	95%	2020-2025	1,373	127	9.25	<0.0001	<0.0001	<0.0001	red
Volatility weighted	95%	2016-2025	2,293	117	5.10	0.822	0.110	0.272	yellow
Volatility weighted	95%	2020-2025	1,373	70	5.10	0.868	0.213	0.454	yellow
Historical simulation	99%	2016-2025	2,293	45	1.96	<0.0001	<0.0001	<0.0001	red
Historical simulation	99%	2020-2025	1,373	44	3.20	<0.0001	<0.0001	<0.0001	red
Volatility weighted	99%	2016-2025	2,293	19	0.83	0.395	0.573	0.594	green
Volatility weighted	99%	2020-2025	1,373	13	0.95	0.842	0.618	0.866	green

Kupiec, Indep. and Joint are p-values for the proportion-of-failures, independence and conditional coverage tests. *Zone* is the worst Basel traffic-light zone reached on any rolling 250-day window, computed at the confidence level of the row rather than at the published 99% boundaries. The book is sized once on 2019, so the 2020–2025 rows are out of sample with respect to sizing.

6.5 Backtest behaviour

The engine's own coverage is reported here because it bears on how much weight the diversification result can carry, and because it replicates a known finding on a new sample. Table 8 gives the four variants.

Plain historical simulation at the disclosed five-year window fails, and fails on count as well as on arrangement: 160 exceptions in 2,293 days is a rate of 6.98% against a nominal 5%, which Kupiec rejects below any conventional level, and the rate rises to 9.25% from 2020. The failures cluster badly — 70 of them

fall in 2022 alone, out of 230 trading days — and the model spends 18.2% of days in the Basel red zone, peaking at 73 exceptions in a 250-day window against a red threshold of 27.

Restating the scenario set in current volatility units repairs it. The Hull and White (1998) variant is not rejected by any of the three tests at either confidence level: at 99% it stays in the green zone throughout, peaking at 4 exceptions in a 250-day window against a yellow threshold of 5. The exception rate falls from 6.98% to 5.10% at 95%, and the independence statistic from 37.7 to 2.6.

This reproduces O'Brien and Szerszeń (2017) on a proxy book and a newer sample. They found bank VaR exceedances excessive and clustered in the crisis, and a conditional-volatility benchmark achieving lower exceedance rates without clustering; the same pattern appears here, with the long disclosed window standing in for the slow adjustment they document.

One comparison must be read carefully. Goldman disclosed 10 exceptions at 95% across the ten years, roughly 0.40% of trading days against a nominal 5%. A literal coverage test on that count would reject calibration at any level, and the rejection would be meaningless, because the two measurements are not the same quantity. Goldman compares VaR against *net revenues*, which include intraday market-making income that is positive far more often than not; the filings say so, and say it is why exceptions are rarer than the confidence level implies. The proxy book has no client flow and no bid–offer spread, so its profit and loss is a clean mark on a static position. The distance between 0.40% and 5% is an indication of how much the revenue cushion is worth, and it is large, but it is not evidence that either model is miscalibrated. The comparison is reported in Appendix A and is not used anywhere in this paper's conclusions.

7 Which assumptions can explain the gap

Section 6 established a gap: the disclosed diversification effect moves by 2.08 percentage points a year and the proxy book's moves by 6.33. This section works through what could account for it, candidate by candidate, and reports which survive.

7.1 Ruled out by magnitude

Two specification choices can be dismissed by size alone, without any argument about mechanism.

The percentile convention is one. Reading the loss quantile by linear interpolation rather than as a discrete order statistic moves average VaR by 0.30% and the mean diversification ratio by 0.06 percentage points. Convexity is the other. Pricing the rate sleeve by full revaluation rather than by a duration approximation moves annual average VaR by between 0.25% and 0.59%, with a largest single-day gap of 1.55%. Neither is within two orders of magnitude of a four-point discrepancy in a ratio's standard deviation.

Both are worth running anyway, because an unexamined choice is an unexamined choice, and because the convexity comparison establishes something useful in its own right: on a one-day horizon with daily yield moves of about five basis points, full revaluation on vanilla bonds buys three tenths of a percent. It earns its cost on instruments whose value is sharply non-linear in the factor, which means options, and on longer horizons. That is a finding about method rather than about Goldman.

7.2 Right direction, not far enough

The scenario window is a real effect and the first candidate that moves the result.

Table 9. Steadiness and coverage move in opposite directions under Goldman's own disclosed settings

Specification	Half-life (days)	Mean (%)	Year-to-year std. dev. (pp)	Exception rate (%)	Kupiec
Lookback window, equally weighted					
252 days (1y)	—	38.2	10.23	5.58	0.209
502 days (2y)	—	39.0	8.87	5.58	0.209
756 days (3y)	—	39.6	8.36	5.89	0.058
1,260 days (5y) †	—	41.6	6.33	6.98	<0.0001
Scenario age weighting, at the disclosed window					
$\lambda = 1.00$ (equal)	—	41.6	6.33	6.98	<0.0001
$\lambda = 0.99$	69	39.9	10.04	5.45	0.328
$\lambda = 0.97$	23	42.0	9.46	5.63	0.177
Goldman, as disclosed	—	42.0	2.08	0.40	—

† marks the disclosed window. The top block holds the weighting equal and varies the window length; the lower block holds the window at the disclosed five years and varies the decay. Reading down the lower block, the year-to-year standard deviation rises while the exception rate falls toward its nominal 5% and Kupiec stops rejecting: adopting Goldman's disclosed weighting makes the engine a better risk model and a worse reproduction of the disclosed steadiness, at the same time. Goldman's own exception rate is measured against net revenues and is not comparable; it is shown for scale only and discussed in Section 6.

The top block of Table 9 runs the window at one, two, three and five years with equal weights. Steadiness improves monotonically as the window lengthens: the year-to-year standard deviation falls from 10.23 points at one year to 8.87 at two, 8.36 at three and 6.33 at the disclosed five. The level climbs toward the disclosure at the same time, from 38.2% to 41.6% against 42.0%.

The direction is right and the destination is not. Five years is the end of this road rather than a point along it, because five years is what Goldman discloses: there is no longer window to appeal to. At the disclosed length the ratio still moves three times as much as the disclosed series, and the engine's coverage has deteriorated badly in the process: the exception rate rises from 5.58% at one year to 6.98% at five, and Kupiec goes from not rejecting to rejecting below any conventional level. A five-year equally weighted scenario set is slow, and 2022 is where that shows: 70 exceptions in 230 trading days, because five years of scenarios in early 2022 reached back to 2017 and contained almost no rate volatility.

7.3 Tested at the disclosed settings, and refuted

The obvious objection to everything above is that a seventeen-factor proxy is a blunter instrument than Goldman's, so the gap measures the replication rather than the bank. The disclosed age weighting tests that objection directly, because it is the one remaining piece of Goldman's stated method that had not been applied, and because it bears on exactly the quantity in dispute: how quickly the engine's view of the correlation structure turns over.

It does not close the gap. It widens it. The lower block of Table 9 holds the window at the disclosed five years and varies the decay. At a 69-day half-life the year-to-year standard deviation rises from 6.33 to 10.04 points; at a 23-day half-life, to 9.46. Against a disclosed 2.08, adopting Goldman's own weighting takes the engine from three times too variable to four or five times too variable.

And it does so while making the engine a better risk model by every coverage test applied to it. The exception rate falls from 6.98% to 5.45% at the longer half-life and 5.63% at the shorter, both close to the

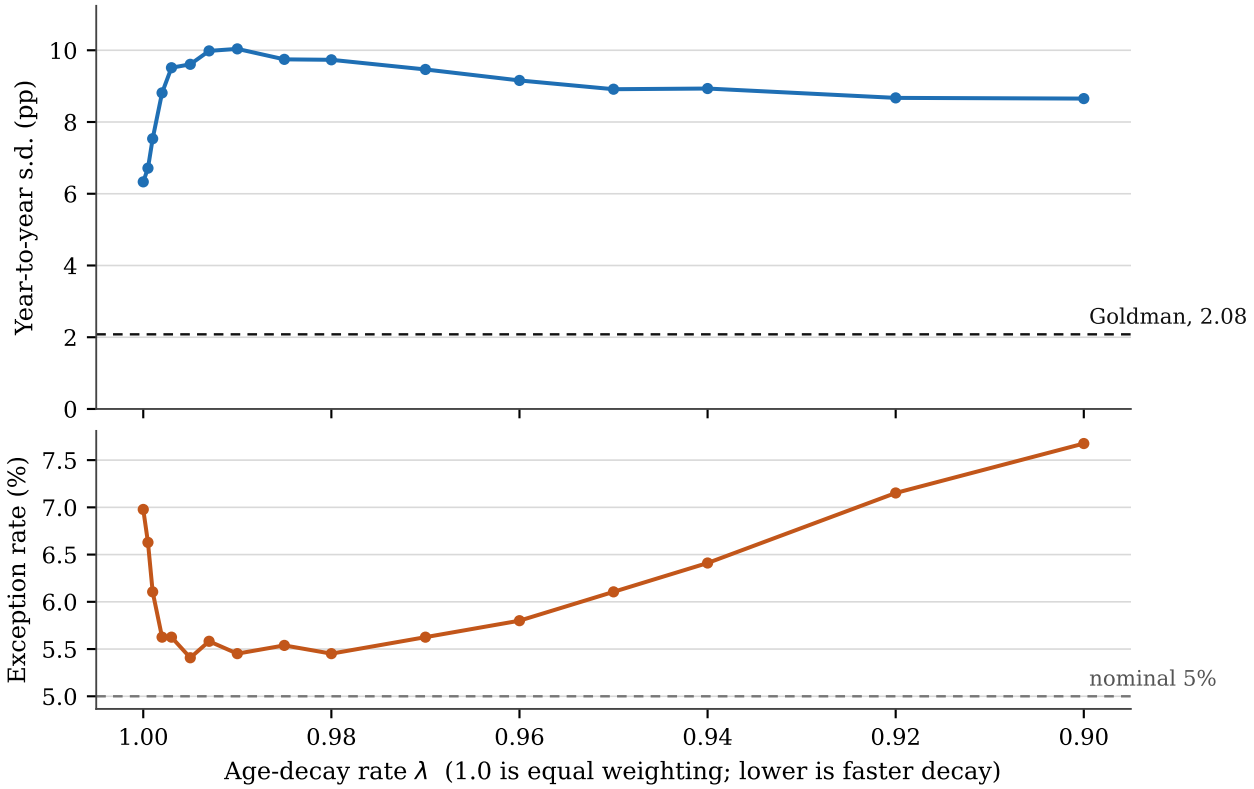


Figure 3. Year-to-year variability and coverage against the age-decay rate, at the disclosed five-year window. Sixteen rates, from equal weighting to a 6.6-day half-life. The upper panel never approaches Goldman's 2.08 anywhere in the grid, and its minimum is equal weighting. The lower panel reaches the nominal 5% exception rate in the middle of the range, where the upper panel is near its worst.

nominal 5%. Kupiec's p-value rises from below 0.0001 to 0.328 and 0.177 respectively, and at the faster decay the independence test passes as well, at 0.305. Weighting toward recent observations is the correct specification for a historical-simulation engine; the filings say so in plain terms, and the backtest agrees with them.

The two movements are in opposite directions and that is the point. Reading down the lower block of Table 9, coverage improves and steadiness degrades together. The gap between the disclosed steadiness and the reproduced steadiness is therefore not an artefact of this replication using a cruder method than the bank. It uses the bank's method, the bank's own window and weighting scheme, and the gap gets larger.

The filings disclose the weighting in kind but not in degree: they say the importance of the data reduces over time without giving the rate. Two rates cannot establish what every rate would do, so the parameter was swept across sixteen values spanning half-lives from 6.6 trading days to 1,386, and Figure 3 reports the result.

The relationship is not monotone, which is worth saying plainly because the obvious argument, that faster decay turns the scenario set over faster and so must increase variability, is wrong. Year-to-year variability rises steeply as decay is introduced, peaks at 10.04 points around a 69-day half-life, and then falls back to 8.65 at the fastest rate tested. The conclusion survives the correction and is now demonstrated rather than asserted: the steadiest configuration anywhere in the grid is equal weighting itself, at 6.33, and every weighted configuration is further from the disclosures than that. The minimum across all sixteen rates is three times Goldman's 2.08.

The lower panel shows the opposing movement across the whole grid rather than at two points. Coverage is best near a 138-day half-life, where the exception rate is 5.41% against a nominal 5% and Kupiec's p-value reaches 0.376, and the year-to-year variability there is 9.61, among the worst in the sweep. There is no rate at which the engine is both well calibrated and as steady as the disclosures, and the sweep is dense enough to say so.

7.4 Bounded by the configuration space

Sleeve direction remains the specification choice with the largest effect on the diversification effect, and Section 6 reported the bound: across all sixteen sign configurations, five are at least as steady as the disclosure and the best of those averages 35.6%, while eight reach 40% or more and the steadiest of those varies by 3.68 points. None does both.

What that bound rules out is a particular kind of objection: that the long-only baseline is simply the wrong corner of the space, and some other corner would land on Goldman. It would have to be a corner that is not in the space. The sixteen configurations exhaust the sign choices available to a four-sleeve book, and the empty quadrant in Figure 2 is a property of all of them.

7.5 What survives

Three candidates the frozen evidence cannot exclude, and they are not ranked, because nothing here ranks them.

The factor count. Seventeen factors against the roughly 70,000 the FY2016 and FY2017 filings describe. More independent sources of risk means more diversification available, so the direction of the bias on the *level* is known. The direction on the *steadiness* is not: a larger factor set might average out regime shifts in any one correlation, or might not. Nothing in public data settles it, and this paper does not guess.

Non-synchronous closes. Tokyo, London and New York do not close at the same instant, which depresses measured cross-market correlation and therefore overstates diversification. This runs opposite to the factor-count bias, which is why neither is used to argue the result in either direction. Neither is quantified.

Table 10 summarises the three verdicts below alongside the exclusions above.

Exposure management. This is the candidate with direct evidence behind it, and Section 6 sharpened what it has to explain. A static book cannot reproduce a path produced by a desk that trades against the market, and Goldman's own commentary says that is what the desk does. In four consecutive filings — FY2021 through FY2024 — the explanation of the year-over-year VaR change has the same shape: volatility moved one way, "partially offset by" exposures moving the other. The firm also moved positions in and out of the measure outright, as the first-quarter 2023 scope change shows. A book whose notionals were fixed in 2019 has no counterpart to any of that.

What Table 7 adds is that this candidate now has a specific job. It does not need to explain why Goldman's ratio moves less than the proxy's, because on a like-for-like basis it does not: period-end against period-end the two have almost the same year-to-year variation. It needs to explain why Goldman's movements revert and the proxy's accumulate. Exposure management is a mechanism that would do exactly that. A desk that sells risk into a correlation regime it dislikes and buys it back when the regime passes produces a ratio that wanders and returns; a fixed book exposed to the same market produces one that wanders and stays. Nothing in the public data establishes that this is what happens, but of the three surviving candidates it is the only one whose mechanism matches the shape of the discrepancy.

Table 10. Candidate explanations for the steadiness gap, and what settles each

Candidate	What settles it	Verdict
Percentile convention	Moves VaR by 0.30% and the mean ratio by 0.06 pp	Excluded
Convexity in the rate sleeve	Full revaluation against duration moves annual VaR by at most 0.59%	Excluded
Lookback window too short	Lengthening 1y→5y cuts the gap from 10.23 to 6.33 pp, against 2.08	Right way, insufficient
Equal weighting, not Goldman's	Swept over 16 decay rates: the steadiest is equal weighting at 6.33 pp and every weighted rate is worse, while coverage improves	Refuted
Wrong sleeve directions	Of 16 configurations, 5 match the steadiness and peak at 35.6%; 8 match the level and bottom out at 3.68 pp	Excluded
Seventeen factors, not 70,000	Not testable on public data; biases the level downward, direction on steadiness unknown	Survives
Non-synchronous closes	Depresses measured cross-market correlation; magnitude not estimated	Survives
Exposures are managed, not held	Goldman attributes VaR changes to exposure moves in four consecutive filings; the only candidate whose mechanism produces reversion rather than drift	Survives

The disclosed year-to-year standard deviation is 2.08 percentage points; every quantity in the middle column is measured against that. *Refuted* is stronger than *excluded*: the disclosed weighting was adopted and moved the gap the wrong way. The three surviving candidates are not ranked, because nothing in the frozen evidence ranks them.

7.6 Taken together

The gap survives the removal of every specification choice that could be removed. It is not the percentile rule, not convexity, not the window length, not the weighting scheme, and not the sleeve signs. What remains is structural: a static four-sleeve book exposed to seventeen public factors, against a trading operation with a far larger factor set that repositions daily.

It is worth restating what the gap is, now that Section 6 has narrowed it. The proxy is not insufficiently variable or excessively variable relative to the disclosures; measured on the same basis its variation is almost identical in size. The proxy's variation persists where Goldman's reverts, and every candidate above has been tested against that discrepancy rather than against a difference in amplitude that does not exist.

Daníelsson et al. (2016) give the mechanism in general form. Model risk, the tendency of different models to disagree, rises with market uncertainty: candidate models agree in calm periods and diverge significantly in distress. Read through the diversification ratio, that is what Section 6 reports. The proxy tracks the disclosure closely through the calm years to 2021, and separates from it in 2022 and after. Their object is risk forecasts in general rather than a disclosed diversification effect, so this is an application of their finding rather than a confirmation of it.

The reading this paper settles on is narrow by design. A static book cannot hold a diversification ratio near 42% through both 2021 and 2023, because the correlation structure beneath it changed and the book did not. Goldman's did hold. Whatever accounts for that is not in the methodology the filings disclose, because the methodology the filings disclose has been applied here and does not produce it.

8 Limitations

Everything in this section constrains how the results should be read. The two that matter most run in opposite directions, and the paper does not claim they cancel.

8.1 Seventeen factors against roughly seventy thousand

This is the largest single difference between the engine here and Goldman's, and it is not fixable from public data. More independent sources of risk means more offsetting available in the joint tail, so a seventeen-factor book should show *less* diversification than a book with a richer factor set. The direction of that bias on the level is therefore known, and it makes the level agreement in Section 6 slightly surprising rather than suspicious.

The direction on the steadiness is not known, and that is the more important gap. A larger factor set might average out regime shifts in any one correlation, which would push toward the disclosed stability; or it might not, if the shifts are common across factors, which is what a macro regime change usually means. Nothing in public data settles it. Section 7 lists it as a surviving candidate and declines to guess, and the central result should be read with that candidate open.

8.2 Non-synchronous closes

Tokyo, London and New York do not close at the same instant. A move that happens in New York after the Tokyo close appears in the Japanese series only the following day, which mechanically depresses measured cross-market correlation and therefore *overstates* diversification.

This runs opposite to the factor-count bias. Neither is quantified here, and the two are not assumed to offset: they are stated so that a reader knows the measured level sits between two unestimated biases of unknown relative size. This is also why neither is used anywhere to argue that the result would move in a particular direction if corrected.

8.3 What the calendar intersection costs

Restricting the panel to days on which every market traded discards about 9% of trading days and leaves 225 to 233 observations a year against the 250 to 253 Goldman reports. The alternative — forward-filling closed markets — would plant runs of exact zeros in the scenario set and understate those factors' risk, which biases the quantity being measured rather than merely reducing the precision of the measurement. The smaller sample is the better of two imperfect options, but it does cost power in every test in Section 5, and the annual figures rest on fewer observations than the disclosures they are compared against.

A related quantisation affects the rate factors. The Treasury curve is quoted to the basis point, so at near-zero policy rates the two-year yield shows no change at all on 21.1% of days across the panel and 43.1% of days in the second half of 2020. That is a real feature of the quoted series rather than a data error, and it slightly understates true rate variability through the near-zero years.

8.4 Instrument substitutions

Commodity exposure runs through exchange-traded funds rather than front-month futures, for the reason given in Section 3. Funds carry roll yield, which drags them away from spot over long horizons. Over the one-day horizon the engine uses it is second order, but it is a real difference from a spot book and it is not corrected.

The rate sleeve mixes US par coupon bonds with euro zero-coupon bonds, each in its source curve's native convention. A zero has longer duration than a par bond of the same maturity, so the two are not

interchangeable per unit of notional. Sizing targets risk rather than notional, which absorbs the difference for the purpose of matching the disclosed split, but the instrument sets are not identical and a reader comparing notionals across sleeves in Table 4 should not read them as comparable exposures.

8.5 What the recalibrated book cannot be used for

The recalibrated book is sized each year using figures Goldman published after that year ended: the 2022 sizing uses a figure released in February 2023. That is appropriate for the structural comparison it is used for, because the question is whether a book with the disclosed risk split generates the disclosed co-movement, and the split is the input. It would be indefensible as a forecast. This is why only the fixed book, sized once on 2019 and never touched again, is backtested, and why the 2020–2025 rows of Table 8 are out of sample with respect to sizing.

8.6 Two limits the disclosed settings impose

Adopting Goldman's stated method closed some latitude and opened a different constraint.

The decay rate behind the disclosed age weighting is not published. The filings say the importance of the data reduces over time without saying how fast, so the scheme is tested in kind but not at Goldman's own rate. The two rates run here were chosen for their half-lives and both move the steadiness the same way, which is what makes the conclusion in Section 7 robust to not knowing the rate, but it remains a conclusion about a family of decay rates rather than about Goldman's.

The window leaves no further room. Five years is what the filings disclose, so the monotone improvement in steadiness with window length cannot be extended by appeal to a longer one. That is a limitation on the argument's escape routes rather than on its evidence: the result is bounded by what the firm says it does.

8.7 What validation does not cover

Finally, and most importantly for interpretation: Section 5 establishes that the engine computes what Section 4 says it computes. It does not establish — and no test on public data could — that the proxy book resembles Goldman's trading book. The level agreement is evidence about the proxy, and the steadiness gap is evidence about what a static proxy cannot do. Neither is a measurement of anything inside the firm.

9 Conclusion

Goldman Sachs discloses one number in Item 7A that depends on something the filing never publishes. The four category VaRs are levels; the diversification effect is a relationship, and it is the only line in the table that a reader outside the firm can test without knowing a single position. This paper tested it, over ten consecutive fiscal years, with a historical-simulation engine run at the bank's own disclosed settings on seventeen public risk factors.

The level transfers. A book sized once on 2019 and then left alone averages 41.6% of summed category VaR against a disclosed 42.0%, a difference no test on ten annual observations distinguishes from zero. That corroborates Pérignon and Smith (2010a), who found disclosed diversification effects close to independently estimated ones on four US banks and quarterly data ending in early 2007. Finding it again on a different bank,

a different decade, a different frequency and a different method is worth reporting precisely because it is not new: a result that survives that much substitution is a result about disclosure practice rather than about one estimation.

The steadiness does not transfer, and that is what this paper adds. Goldman's average daily ratio has a year-to-year standard deviation of 2.08 percentage points across a decade containing a pandemic, a zero-rate period and more than 500 basis points of policy tightening. The proxy's is 6.33, a difference a variance-ratio test puts at $p = 0.0028$, and the two move oppositely over the sample: Goldman's second-half mean is 0.8 points above its first, the proxy's 8.5 below.

The firm's second disclosed measure says what kind of failure it is. Goldman also publishes a period-end snapshot of the same ratio, and that series has a standard deviation of 6.72 points. Its diversification effect moves a great deal, and what is stable is the annual mean rather than the quantity. Measured period-end against period-end, the proxy's variation cannot be distinguished from Goldman's. What distinguishes them is persistence: the lag-1 autocorrelation of consecutive annual values is −0.47 for Goldman's average daily series and about zero for its period-end series, against roughly +0.9 for both proxy series. Goldman's movements do not carry forward; the proxy's do. The proxy reproduces the level, and its variation is the same size as the disclosed variation; what it fails to reproduce is the mean reversion, and the mean reversion is what the disclosed steadiness consists of.

What makes the result hard to dismiss is that it survives being handed the bank's own method. The filings disclose a five-year scenario window and age-weighted scenarios, and both were implemented. The window moves the gap the right way and runs out of room, since five years is what is disclosed and there is no longer window to appeal to, and the weighting moves it the wrong way, raising the standard deviation to between 9.46 and 10.04 points while simultaneously repairing the engine's coverage, from a 6.98% exception rate that Kupiec rejects below 0.0001 to 5.45% at $p = 0.328$. The engine gets better as a risk model and worse as a reproduction of the disclosed steadiness, in the same move. Whatever holds Goldman's ratio still is not in the methodology the filings describe, because that methodology has been applied here and does not produce it.

Nor is it a matter of having guessed the sleeve directions wrongly. Across all sixteen sign configurations the two properties trade off: five are at least as steady as the disclosure and the best of those averages 35.6%, eight reach 40% or more and the steadiest of those varies by 3.68 points. None does both. That trade-off, rather than any single calibration, is what the sign space establishes. It is a descriptive bound on what this engine can produce rather than a test — ten annual observations per configuration will not support one — and it is what an outside party can establish from public data.

Taken with the decomposition above, the paper's contribution is two-part: a replication run at the bank's own disclosed settings, which rules out the objection that the gap measures a cruder method; and a separation of amplitude from persistence using a second disclosed series the firm already publishes, which identifies what the gap actually consists of. Neither requires anything the filings do not contain.

How much this establishes, and how much it does not

The claim is narrower than the result might be made to sound. This paper does not show that Goldman's disclosed diversification effect is implausible, managed, or anything other than what the firm says it is. It shows that a static four-sleeve book of public factors cannot reproduce it, under the method the firm discloses, across the full space of sign configurations available to such a book. Those are different statements, and the second does not imply the first.

Three explanations survive the elimination in Section 7, and the paper does not rank them because nothing in the evidence ranks them. The factor count, seventeen against roughly seventy thousand, is the largest structural difference and its effect on steadiness is simply unknown. Non-synchronous closes across time zones bias the measurement in the opposite direction to the factor count, and neither bias is quantified. And exposures are managed rather than held: in four consecutive filings, FY2021 through FY2024, Goldman attributes its year-over-year VaR change to volatility moving one way and exposures moving the other. A book whose notionals were fixed in 2019 has no counterpart to that. It is the one surviving candidate with direct documentary support, and the only one whose mechanism matches the shape of the discrepancy: a desk that sheds risk into an unwelcome correlation regime and restores it afterwards produces a ratio that returns to where it was, which is precisely what the proxy's does not do.

What would settle it

Separating those three requires data that is not public. Position-level or desk-level exposures would do it directly: holding the factor set fixed and varying only whether positions are repriced or held would isolate exposure management from the other two, and a richer factor set run against the same book would isolate the factor count. Supervisory data of the kind O'Brien and Szerszeń (2017) worked with would get part of the way, though it carries the trading outcomes rather than the diversification decomposition this question needs.

Absent any of that, the bound is what an outside party can establish, and establishing it is the point. A reader of a 10-K can now say something specific about the one line in the market-risk table that is testable from outside: its level is reproducible from public data, the size of its variation is reproducible too, and its persistence is not, under the bank's own disclosed window and weighting, swept across the range the filings leave open.

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Data availability statement

Every input to this study is public. The ten Form 10-K filings are listed with their accession numbers in Table A.1 and remain retrievable from EDGAR; the Market Risk Management section of each is preserved verbatim under `references/sec_filings/`, which is the text every disclosed figure was read from, and the parsed series are tracked as CSV so the disclosed side of every comparison can be inspected without running anything.

The market data behind the factor panel cannot be preserved the same way. The US Treasury par yield curve is a work of the US government and the European Central Bank's euro area curve is openly licensed

with attribution; either could be redistributed. The exchange-traded closes come from a commercial endpoint whose terms do not permit redistribution, and because the aligned panel interleaves all three sources, no part of it is tracked. What is tracked is `scripts/02_build_factors.py`, which rebuilds the panel from the same public endpoints over the window recorded in `config/params.yaml`. A reader reproducing this paper regenerates the factor data rather than receiving it.

Code availability

The disclosure retrieval and parser, the factor construction, the Value-at-Risk engine, the calibration, the backtest, the regime and robustness analyses and the test suite described in Section 5 are all available in the repository. Every table and figure in this paper is generated from the tracked analysis outputs by a script under the manuscript's own `build/` directory; none is transcribed by hand, and deleting the generated files and rerunning those scripts reproduces the manuscript unchanged. One exception is worth naming: Table 3 reads the aligned factor panel for the sentence recording how many trading days it spans, and that panel is the one input not tracked, for the reason given above. Rebuilding it restores the table; skipping it costs that table and nothing else. The commit the reported results were produced from, together with a SHA-256 digest of each analysis output they depend on, is recorded under `audit/`, so a reader can confirm that a given number corresponds to the state of the code that produced it.

Declaration of generative AI use

During the preparation of this work, the author used Claude in order to revise and restructure the manuscript, prepare the plotting script, literature checks and LaTeX preparation. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Table A.1. The filings the disclosed series is read from

Fiscal year	Period end	Filed	Accession number
FY2016	2016-12-31	2017-02-27	0001193125-17-056804
FY2017	2017-12-31	2018-02-26	0001193125-18-056383
FY2018	2018-12-31	2019-02-26	0001193125-19-050198
FY2019	2019-12-31	2020-02-21	0001193125-20-043853
FY2020	2020-12-31	2021-02-22	0001193125-21-049380
FY2021	2021-12-31	2022-02-25	0001193125-22-052682
FY2022	2022-12-31	2023-02-24	0000886982-23-000003
FY2023	2023-12-31	2024-02-23	0000886982-24-000006
FY2024	2024-12-31	2025-02-27	0000886982-25-000005
FY2025	2025-12-31	2026-02-25	0000886982-26-000091

The Goldman Sachs Group, Inc., Annual Report on Form 10-K, CIK 0000886982. Each is retrievable from EDGAR at `https://www.sec.gov/Archives/edgar/data/886982/<accession>/`. Every figure in this paper is taken from the Market Risk Management section of one of these ten documents.

Table A.2. Period-end Value-at-Risk by risk category, FY2016–FY2025

	Risk category							
Year	Rates	Equities	FX	Commod.	Summed	Diversif.	Total	% of sum
2016	45	34	23	19	121	-39	82	32.2
2017	48	31	7	9	95	-30	65	31.6
2018	46	32	12	11	101	-44	57	43.6
2019	54	24	10	10	98	-28	70	28.6
2020	60	50	11	16	137	-46	91	33.6
2021	69	31	19	30	149	-58	91	38.9
2022	108	27	35	18	188	-85	103	45.2
2023	93	25	15	14	147	-54	93	36.7
2024	84	41	38	14	177	-86	91	48.6
2025	59	45	17	18	139	-60	79	43.2

In millions of US dollars, except the final column. The analysis in this paper uses average daily VaR (Table 1); the period-end series is a single-day snapshot and is reported here for completeness. The identity holds in this series too.

A The disclosed series in full

Table A.1 identifies the ten filings every disclosed figure in this paper is read from. Accession numbers are given so that any figure can be traced to the document it came from and retrieved independently.

Table 1 in Section 2 gives the average daily series, which is what the analysis uses. Table A.2 gives the period-end series for completeness. The two measure different things: the average daily figure summarises a year of risk-taking, while the period-end figure is a single day's snapshot and is correspondingly noisier. The diversification identity holds in both.

Section 6 uses the period-end series to separate the amplitude of the disclosed variation from its persistence, and Table 7 carries that comparison. The point in brief: measured period-end Goldman's own ratio has a year-to-year standard deviation of 6.72 percentage points against 2.08 for the average daily series, so the disclosed steadiness is a property of the annual mean rather than of the ratio itself.

Table A.3. Exception counts at 95%: the proxy book against Goldman's disclosed counts

	Proxy book, plain historical simulation			Goldman
Year	Days	Exceptions	Rate (%)	Exceptions
2016	233	9	3.9	0
2017	232	5	2.2	0
2018	230	11	4.8	2
2019	225	8	3.6	0
2020	227	12	5.3	2
2021	231	11	4.8	1
2022	230	70	30.4	2
2023	233	19	8.2	1
2024	228	11	4.8	2
2025	224	4	1.8	0
Total	2,293	160	6.98	10

These two columns do not measure the same thing and should not be compared as though they did. Goldman scores VaR against daily net revenues, which include intraday market-making income that is positive far more often than not; the proxy book has no client flow and no bid–offer spread, so its profit and loss is a clean mark on a static position. The table is reported for completeness and the gap is an indication of what the revenue cushion is worth, not evidence that either model is miscalibrated. No conclusion in this paper rests on it.

The disclosed high and low VaR by category point the same way. Taken as first reported, the median within-year ratio of high to low runs from 1.9 for interest rates to 4.3 for currency rates. A book whose category VaRs move through ranges that wide over a year is not a book whose diversification effect sits still from day to day, and the filings are explicit enough about this that no inference is needed.

Exception counts

Section 6 defers this comparison here because it cannot carry weight. Table A.3 gives it.

Goldman recorded 10 exceptions at the 95% risk-management measure across the ten years, against 160 for the proxy book. The gap is not evidence about either model's calibration, because the two are scored against different quantities: Goldman compares VaR against daily net revenues, which include intraday market-making income that is positive on the large majority of days, while the proxy book is a static position marked cleanly with no client flow and no bid–offer spread. The filings make this point themselves, observing that in periods with far more positive than negative net revenue days, fewer exceptions are expected than the confidence level implies.

The distance between 0.40% and 6.98% of days is therefore best read as an indication of how much the revenue cushion is worth, and it is large. It is not used anywhere in this paper's conclusions.

B Engine specification

This appendix records three implementation details that Section 4 states but does not develop.

Reading the quantile off a finite scenario set

With $W = 1{,}260$ scenarios and $\alpha = 0.05$, the 5% point falls between the 63rd and 64th worst outcome, and the convention for reading it is not disclosed. Two are implemented. Linear interpolation between the bracketing order statistics is the default. The discrete reading takes the $\lceil \alpha W \rceil$th worst outright, which is the more conservative of the two and what a bank wishing to avoid understating risk would choose.

The ceiling matters rather than the floor. Flooring agrees with the ceiling whenever αW is fractional and silently picks the wrong side whenever it is a whole number, which is precisely the case a unit test is likely to be written around. Section 7 reports the difference between the two conventions: 0.30% of the VaR level and 0.06 percentage points of the mean diversification ratio.

Under age weighting the same two conventions carry over, with cumulative weight in place of cumulative count. The weighted interpolation uses the plotting position $(F_j - w_j)/(1 - w_j)$, where F_j is cumulative weight through the jth ordered outcome. Under equal weights this reduces to $(j - 1)/(W - 1)$, which is what the unweighted percentile interpolates against, so the two estimators coincide exactly at $\lambda = 1$. That identity is what Section 5 pins to machine precision, and it is the reason the age-weighting results in Section 7 can be read as effects of the weighting rather than of a change of estimator.

Why the sleeve calibration is closed form

Every position in the proxy book has profit and loss proportional to its notional: a linear position returns the factor's return times notional, and a bond position returns the revaluation ratio of Equation (1) times notional. Scaling a sleeve's notionals by k therefore scales every scenario outcome for that sleeve by k, which scales the order statistics by k, which scales the sleeve's VaR by k on every date and hence its average over any period. Equation (2) follows, and one division per sleeve lands on the disclosed target exactly.

The property is positive homogeneity, the third of the four axioms in Artzner et al. (1999). No iteration or search is involved, which is why Table 5 reports a residual at the scale of floating-point noise rather than a convergence tolerance. Had the book contained options or any other position whose value is non-linear in notional, the argument would fail and the calibration would need solving numerically.

The look-ahead probe

The timing convention — VaR dated t from scenarios up to t, scored against the outcome on $t + 1$ — is enforced by a test rather than by inspection, because look-ahead is the one error class that makes a model look better rather than worse.

The probe multiplies every factor move after a cut-off date by 25 and re-runs the engine on the tampered panel. Every column dated on or before the cut-off must be identical to its untampered value. The realised profit-and-loss column is excluded, because that column is supposed to change: it carries the outcome of the following day, which is exactly what the tampering alters. A separate test confirms that the realised result recorded against date t is the book repriced on the move of $t + 1$ rather than of t, which is the off-by-one that would make the backtest meaningless while leaving every number plausible.

Table C.1. The fixed book across market regimes

Regime	Days	Mean VaR	Diversif. (%)	Realised P&L s.d.	Exception rate (%)	Equity vol (ann., %)	10y vol (bp/day)
Calm, 2017–2019	687	54.8	44.8	28.7	3.5	12.7	4.0
COVID shock, 2020	82	56.8	45.4	67.9	11.0	57.2	8.8
Recovery, 2021	231	52.9	45.8	31.0	4.8	13.7	4.5
Rate shock, 2022	195	64.9	42.3	94.9	32.3	25.4	8.4
After the shock, 2024–2025	452	99.7	32.1	49.2	3.3	16.7	5.8

The book is sized once on 2019 and never changed, so every difference across rows is the market rather than the book. Mean VaR and realised profit-and-loss standard deviation are in millions of US dollars at the book's normalising scale and carry no information in levels; their ratio within a row does. The last row is where the downward drift in the measured diversification effect shows up, with VaR at its sample peak and volatility back to ordinary levels.

C The fixed book across regimes

Table C.1 cuts the fixed book's daily output into five market regimes. Because the book is sized once on 2019 and never changed, every difference between rows is the market rather than the book, which makes this the cleanest view of how a static portfolio behaves through a regime change.

Three rows carry the argument of Section 7.

The COVID window shows the lag. Equity volatility ran at 57.2% annualised and realised profit and loss had a standard deviation of 67.9, while VaR averaged 56.8 — barely above its calm-period level of 54.8 — because the shock was six weeks old and the five-year scenario set was still overwhelmingly quiet data. Roughly one day in nine breached.

The 2022 rate shock shows the same mechanism at its worst, and it is the clearest cost of running the disclosed window without the disclosed weighting. The exception rate reaches 32.3%. Five years of scenarios in early 2022 reach back to 2017, a stretch containing almost no rate volatility, so the engine entered the fastest tightening cycle in four decades carrying a scenario set that had never seen one.

The final row is where the downward drift in the measured diversification effect appears. Mean VaR sits at its sample peak of 99.7 with volatility back to ordinary levels and the exception rate collapsed to 3.3%, and the diversification effect has fallen to 32.1% from 44.8% in the calm years. The book did not change. The window did: it now contains 2022 and will continue to until 2027. That ratchet is what Section 6 identifies as the source of the trend in the model series, and it is the behaviour a dealer repositioning daily does not exhibit.